Blanding’s Turtle

Which restoration strategy should be implemented?

Course: Grade 12 Data Management (MDM4U)
Format: Evidence-based recommendation using summary statistics + graphs

https://canadianherpetology.ca/species/species_page.html?cname=Blanding%27s%20Turtle

Establishing Context

Problem Overview

- Blanding's turtle is an endangered species in Ontario that depends on healthy wetlands and safe movement between them. Habitat loss and fragmentation reduce nesting success and increase adult deaths, especially from roads. The goal of restoration is to increase long term population survival by improving habitat quality and reducing major threats.

Strategies

- Strategy 1 Habitat Expansion
  This strategy increases the total amount of suitable wetland habitat by restoring or creating wetlands and improving nearby land. It aims to support more food, shelter, and nesting areas so more turtles can survive and reproduce.
- Strategy 2 Habitat Connectivity
  This strategy connects separate wetland areas using protected corridors (for example vegetated pathways or wildlife crossings). It aims to reduce fragmentation so turtles can safely move between habitats to access nesting sites and resources, and to support a more stable population over time.

https://giving.natureconservancy.ca/product/blandings-turtle/

1) What is the task?

Goal

- Compare two restoration strategies for Blanding's turtles using the provided dataset.
- Use statistics and graphs to decide which strategy is more effective and sustainable.
- Explain your recommendation so a non-expert can understand.

How I will do it

- Choose variables (population, nest success, adult mortality, cost, stability).
- Calculate summary statistics (mean, median, range, IQR, standard deviation).
- Create at least two graphs and interpret the trends (short vs long term).
- Identify trade-offs and real-world constraints, then recommend a strategy.

2) Strategies being compared

Habitat Expansion

- Increase protected wetland + upland habitat area.
- More space = more nesting and feeding sites.
- Often paired with road-risk reduction around key areas.

Key “extra” variable in dataset:

Road mortality (count)

Habitat Connectivity

- Build wildlife corridors to link fragmented habitats.
- Helps movement for nesting, feeding, and overwintering.
- Dataset includes “Connected” vs “Isolated” habitats.

Key “extra” variables in dataset:

Corridor length (km) + population variability (±)

3) Understanding the dataset

- Each strategy has its own dataset (based on how similar projects worked at other sites).
- The years start at 2012 for both so we can compare trends over the same time window.
- The starting populations are different because the sites (or starting conditions) were different.
- To compare fairly, focus on change over time (increase/decrease and % change), not only the starting number.
- Connectivity is measured every 2 years, so graphs compare the shared years (2012, 2014, …, 2022).

Key idea:

Use the same variables + the same time period to make a fair comparison.

4) Variables used to measure effectiveness

Main outcome (bigger is better)

- Estimated population (turtles)
  - A higher estimated population is better because it lowers extinction risk and shows the habitat is supporting overall survival and reproduction over time.

Supporting indicators

- Nest success rate (%) — higher is better (more hatchlings).
- Adult mortality rate (%) — lower is better (adults are slow to replace).
- Cost ($k) — lower / slower growth is better for long-term funding.
- Stability — lower variability (±) suggests more reliable, stable outcomes.
- Road mortality (Expansion dataset) — lower is better (direct adult survival).

5) Summary statistics — Habitat Expansion (2012–2022)

Variable	Mean	SD	Range	IQR
Estimated population	452.1	17.9	55	33
Nest success (%)	54.0	3.0	10.0	4.0
Adult mortality (%)	5.14	0.55	1.70	0.90
Cost ($k)	154.1	22.9	65	45
Road mortality (count)	31.3	5.3	16	9

Interpretation: higher average population + lower road mortality suggests stronger survival benefits.

6) Summary statistics — Habitat Connectivity (Connected habitat only)

Variable	Mean	SD	Range	IQR
Estimated population	441.8	19.4	50	33
Nest success (%)	54.2	3.5	9.0	6
Adult mortality (%)	5.08	0.45	1.20	0.70
Cost ($k)	165.0	24.7	65	40
Population variability (±)	27.5	3.1	8	5
Corridor length (km)	28.3	12.4	33	20

Interpretation: variability (±) decreases over time, suggesting a more stable connected population.

7) Worked examples (showing calculations)

Example A: Mean (Expansion estimated population, 2012–2022)

Data: 420, 428, 435, 448, 455, 452, 460, 468, 462, 470, 475
Sum = 4973
Mean = 4973 ÷ 11 = 452.1 turtles

Example B: Median (same dataset)

There are 11 values, so the median is the 6th value when ordered → 455 turtles.

Example C: Range (Expansion road mortality)

Max = 41, Min = 25 → Range = 41 − 25 = 16 road deaths

I included sample calculations to demonstrate my process and verify the accuracy of my results.

8) Worked examples (IQR and standard deviation)

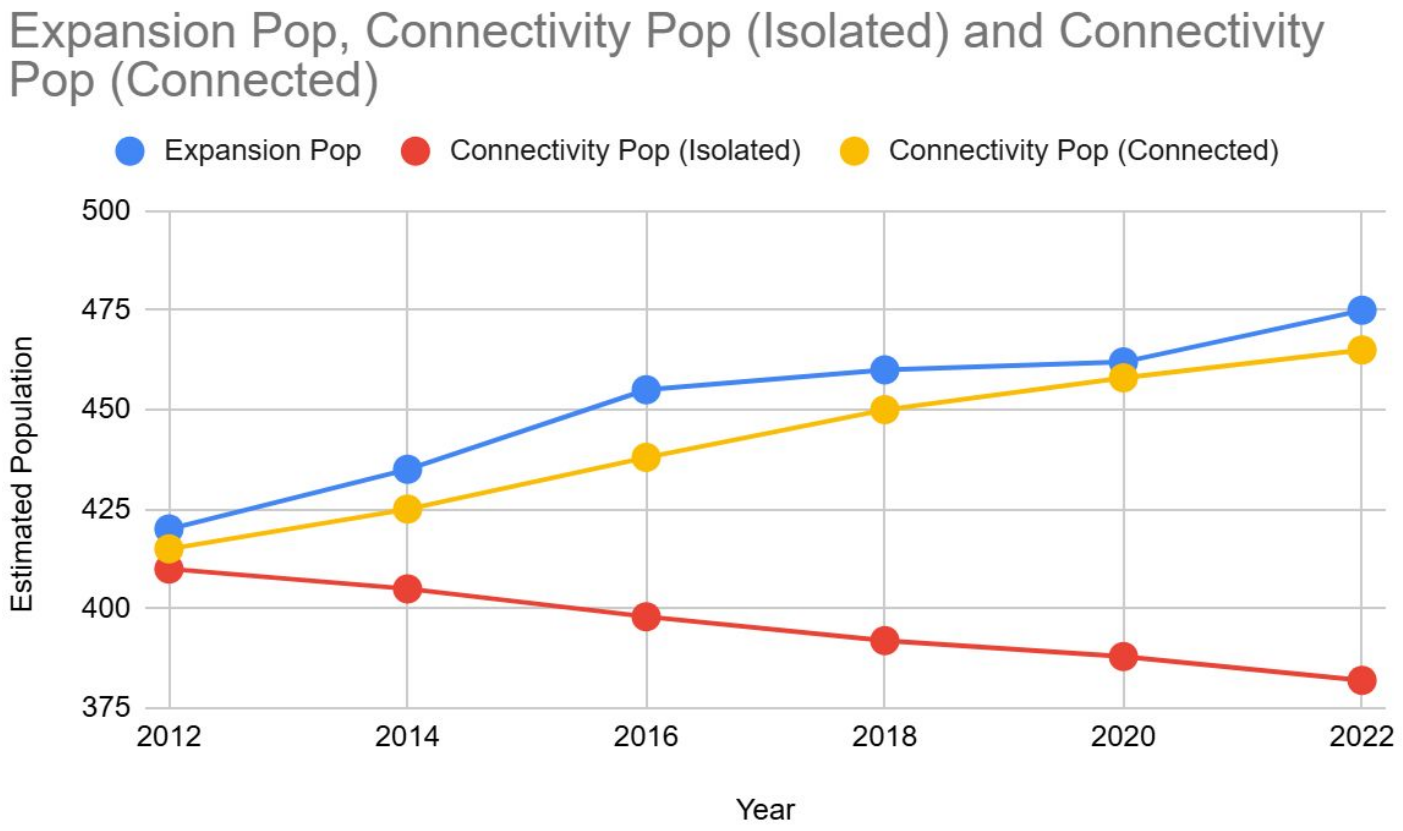
Example D: IQR (Expansion estimated population)

Ordered data: 420, 428, 435, 448, 452, 455, 460, 462, 468, 470, 475
Exclude the median (455).
Lower half: 420, 428, 435, 448, 452 → Q1 = 435
Upper half: 460, 462, 468, 470, 475 → Q3 = 468
IQR = Q3 − Q1 = 468 − 435 = 33 turtles

Example E: Standard deviation (what it means)

SD measures typical distance from the mean.
For Expansion population, SD ≈ 17.9 turtles, meaning most yearly estimates are about 18 turtles from the mean (452.1).
Lower SD → more consistent estimates over time.

9) Graph 1 — Estimated population over time (shared years)



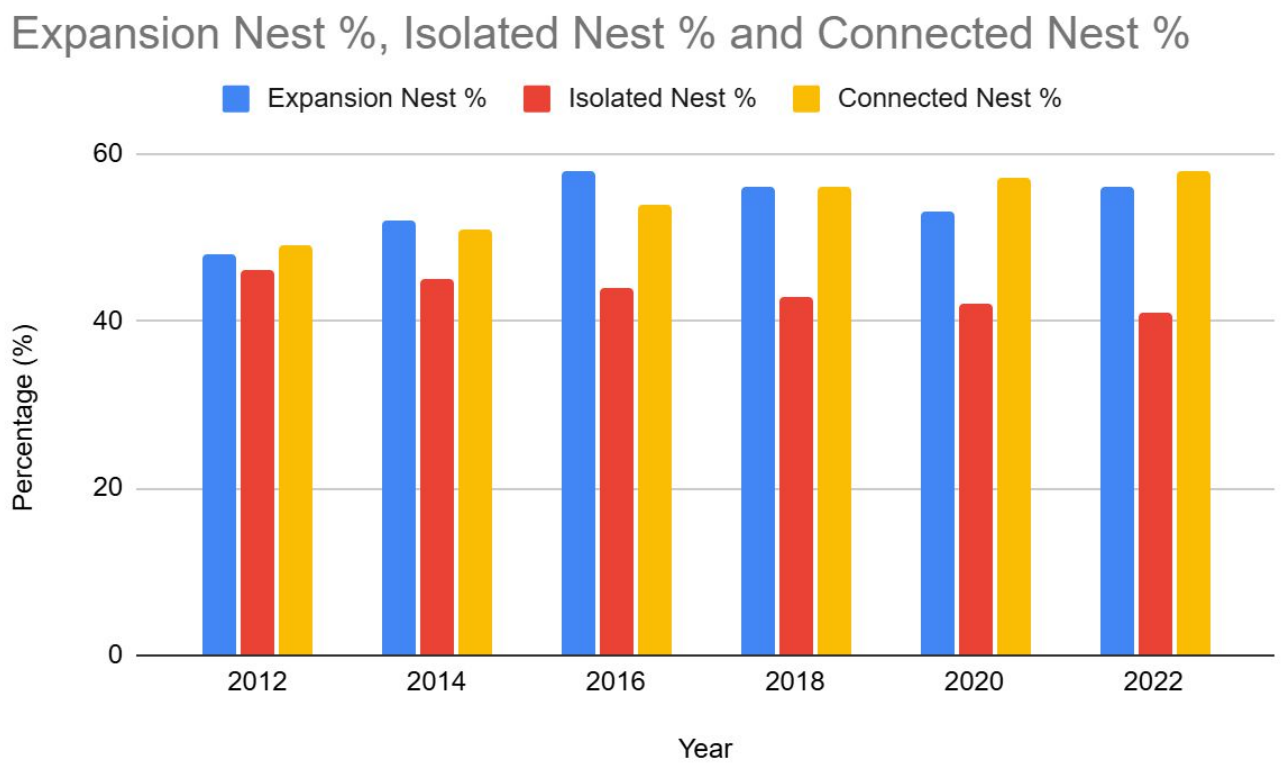
What the graph shows

- Both lines go upward → both strategies help the population grow.
- Expansion: 420 → 475 (+55, 13.1%).
- Connectivity: 415 → 465 (+50, 12.0%).
- Early growth (2012→2016): Expansion +35 vs Connectivity +23.
- So, expansion grows faster early and ends slightly higher overall.
- We only used the shared years (every 2 years) to keep the comparison fair.

Why a line graph?

It's the best way to see trends (up/down) and compare slopes over time.

10) Graph 2 — Nest success rate comparison (shared years)



What the graph shows

- Nest success = % of nests that produce hatchlings (higher is better).
- Both strategies improve overall from ~50% to the high-50s.
- 2016: Expansion is higher (58%) than Connectivity (54%).
- 2020 and 2022: Connectivity is slightly higher (57% vs 53%, then 58% vs 56%).
- This suggests connectivity may help reproduction a bit more in later years.
- But the difference is small, so I also compared adult survival + road deaths.

Why clustered bars?

They let you compare the two strategies side-by-side for the same year.

13) Interpreting the evidence (short-term vs long-term)

Short term (2012 → 2016)

- Expansion: Population rises from 420 to 455 (+35, +8.3%), nest success rises from 48% to 58% (+10 points), and adult mortality drops from 6.2% to 4.9% (−1.3 points) while cost increases from $120k to $150k (+$30k). This means expansion shows stronger early improvement in both population growth and adult survival indicators, and it is more cost effective early at about $857 per turtle gained ($30,000 ÷ 35).
- Connectivity (Connected): Population rises from 415 to 438 (+23, +5.5%), nest success rises from 49% to 54% (+5 points), and adult mortality drops from 5.8% to 5.1% (−0.7 points) while cost increases from $130k to $160k (+$30k). This means connectivity helps, but the early gains are smaller for the same cost increase, and the cost effectiveness is lower at about $1,304 per turtle gained ($30,000 ÷ 23).
- Conclusion: In the short term, expansion is the stronger strategy because it produces larger population increases and larger improvements in adult survival measures for the same additional cost, which suggests faster early recovery.

Long term (2012 → 2022)

- Expansion: Population rises from 420 to 475 (+55, +13.1%), adult mortality drops from 6.2% to 4.5% (−1.7 points), and road deaths drop from 41 to 25 (−16) while cost rises from $120k to $185k (+$65k). This matters because reducing adult deaths, especially road deaths, is critical for turtles that take many years to mature, and the long term cost effectiveness is about $1,182 per turtle gained ($65,000 ÷ 55).
- Connectivity (Connected): Population rises from 415 to 465 (+50, +12.0%), nest success rises from 49% to 58% (+9 points), and variability drops from 32 to 24 (−8) while cost rises from $130k to $195k (+$65k). This suggests connectivity supports reproduction and makes the population estimates more stable over time, but the evidence for improved adult survival is less direct, and cost effectiveness is about $1,300 per turtle gained ($65,000 ÷ 50).
- Conclusion: Long term, both strategies work, but expansion is the better overall choice because it shows stronger adult survival evidence through lower road mortality and a slightly larger population increase, while also being slightly more cost effective per turtle gained.

14) Trade-offs, constraints, and final recommendation

Trade-offs / constraints

- Cost: connectivity tends to be slightly higher in shared years.
- Feasibility: expansion needs land protection; corridors need cooperation across land/roads.
- Time: both take years; connectivity data is every 2 years (less detail).
- Different “benefits”: expansion reduces road deaths; connectivity improves movement and stability.

Recommendation

Implement: Habitat Expansion

- Largest long-term population increase (+55 turtles).
- Strongest drop in adult mortality (−1.7 percentage points).
- Only dataset showing large drop in road mortality (−16 road deaths).
- Similar cost growth, and usually slightly cheaper than connectivity in shared years.
- Connectivity can still be a “Phase 2” add-on once key habitat is protected.

15) Data quality + next steps + reflection

Strengths of the dataset

- Multiple variables (not just population) gives a fuller comparison.
- Multi-year data makes trends clearer than a single-year snapshot.

Limitations / fairness issues

- Expansion has yearly data; connectivity has every-2-years data (less detail).
- Not a controlled experiment—weather, predators, and human activity could also affect results.
- Unaware of how the data was collected / unsure of the method at which data was collected

Follow-up investigation (improves decisions)

- Collect yearly connectivity data (to match the expansion timeline).
- Add juvenile survival (hatchling → adult) to connect nest success to long-term population growth.
- Track road-crossing hotspots and traffic volume to explain road mortality changes.

Final decision

Habitat Expansion is the best overall strategy for population growth + adult survival.

Tip: Put your name, teacher, and due date in the title slide before submitting.